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Transactions and Concurrency

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Marks 1 14 questions

Q1. GATE CSE 2026 Set 2

Consider concurrent execution of two transactions $T1$ and $T2$ in a DBMS, both of which access a data object A. For these two transactions to not conflict on A, which one of the following statements must be true?

(A) Both $T1$ and $T2$ only read A
(B) $T1$ reads A and $T2$ writes A
(C) $T1$ writes A and $T2$ reads A
(D) Both $T1$ and $T2$ write A

Open solution

Q2. GATE CSE 2025 Set 2

An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account A attempted simultaneous transfers of Rs. 10000 each from account A to account B. Both transactions read the same value, Rs. 11000, as the initial balance in A and were allowed to go through. B was credited Rs. 10000 twice. A was debited only once and ended up with a balance of Rs. 1000. Which of the following properties is/are certain to have been violated by the system?

(A) Atomicity (B) Consistency (C) Isolation (D) Durability

Open solution

Q3. GATE CSE 2025 Set 1

A schedule of three database transactions T_1, T_2, and T_3 is shown. $R_i(A)$ and $W_i(A)$ denote read and write of data item A by transaction $T_i, i = 1, 2, 3$. The transaction T_1 aborts at the end. Which other transaction(s) will be required to be rolled back?

$R_1(X)W_1(Y)R_2(X)R_2(Y)R_3(Y)\ \text{ABORT}\ (T_1)$

(A) Only T_2
(B) Only T_3
(C) Both T_2 and T_3
(D) Neither T_2 nor T_3

Open solution

Q4. GATE CSE 2024 Set 2

Once the DBMS informs the user that a transaction has been successfully completed, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called

(A) durability (B) atomicity (C) consistency (D) isolation

Open solution

Q5. GATE CSE 2024 Set 2

Which of the following statements about the Two Phase Locking (2PL) protocol is/are TRUE?

(A) 2PL permits only serializable schedules
(B) With 2PL, a transaction always locks the data item being read or written just before every operation and always releases the lock just after the operation
(C) With 2PL, once a lock is released on any data item inside a transaction, no more locks on any data item can be obtained inside that transaction
(D) A deadlock is possible with 2PL

Open solution

Q6. GATE CSE 2021 Set 1

Suppose a database system crashes again while recovering from a previous crash. Assume checkpointing is not done by the database either during the transactions or during recovery.

Which of the following statements is/are correct?

(A) The same undo and redo list will be used while recovering again.
(B) The database will become inconsistent.
(C) All the transactions that are already undone and redone will not be recovered again.
(D) The system cannot recover any further.

Open solution

Q7. GATE CSE 2019

Consider the following two statements about database transaction schedules:

I. Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable.

II. Timestamp-ordering concurrency control protocol with Thomas Write Rule can generate view serializable schedules that are not conflict serializable.

Which of the above statements is/are TRUE?

(A) Both I and II (B) I only (C) II only (D) Neither I nor II

Open solution

Q8. GATE CSE 2016 Set 2

Suppose a database schedule S involves transactions $T_1, \ldots, T_n$. Construct the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?

(A) Topological order
(B) Depth-first order
(C) Breadth-first order
(D) Ascending order of transaction indices

Open solution

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Q9. GATE CSE 2016 Set 1

Which one of the following is **NOT** a part of the $ACID$ properties of database transactions?

(A) Atomicity
(B) Consistency
(C) Isolation
(D) Deadlock-freedom

Open solution

Q10. GATE CSE 2015 Set 2

Consider the following transaction involving two bank accounts x and y.

```
read (x) ; x: = x - 50; write (x); read (y); y: = y + 50;   write (y)
```

The constraint that the sum of the accounts x and y should remain constant is that of

(A) Atomicity
(B) Consistency
(C) Isolation
(D) Durability

Open solution

Q11. GATE CSE 2010

Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?
I. 2-phase locking
II. Time-stamp ordering

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Open solution

Q12. GATE CSE 2005

Amongst the ACID properties of a transaction, the 'Durability' property requires that the changes made to the database by a successful transaction persist

(A) Except in case of an Operating System crash
(B) Except in case of a Disk crash
(C) Except in case of a power failure
(D) Always, even if there is a failure of any kind

Open solution

Q13. GATE CSE 2004

Which level of locking provides the highest degree of concurrency in a relational database?

(A) Page.
(B) Table.
(C) Row.
(D) Page, table and row level locking allow the same degree of concurrency.

Open solution

Q14. GATE CSE 2003

Which of the following scenarios may lead to an irrecoverable error in a database system?

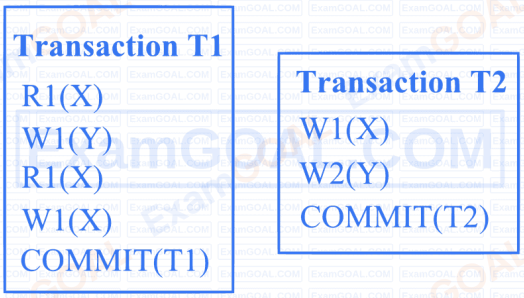
(A) A transaction writes a data item after it is read by an uncommitted transaction
(B) A transaction reads a data item after it is read by an uncommitted transaction
(C) A transaction reads a data item after it is written by a committed transaction
(D) A transaction reads a data item after it is written by an uncommitted transaction

Open solution

Marks 2 23 questions

Q15. GATE CSE 2025 Set 2

Consider the database transactions T 1 and T 2, and data items X and Y . Which of the schedule(s) is/are conflict serializable?



(A) $R1(X), W2(X), W1(Y), W2(Y), R1(X), W1(X), COMMIT(T2), COMMIT(T1)$
(B) $W2(X), R1(X), W2(Y), W1(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)$
(C) $R1(X), W1(Y), W2(X), W2(Y), R1(X), W1(X), COMMIT(T1), COMMIT(T2)$
(D) $W2(X), R1(X), W1(Y), W2(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)$

Open solution

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Q16. GATE CSE 2024 Set 1

Consider the following read-write schedule S over three transactions T_1, T_2, and T_3, where the subscripts in the schedule indicate transaction IDs:

$S : r_1(z); w_1(z); r_2(x); r_3(y); w_3(y); r_2(y); w_2(x); w_2(y);$

Which of the following transaction schedules is/are conflict equivalent to S?

(A) $T_1T_2T_3$ (B) $T_1T_3T_2$ (C) $T_3T_2T_1$ (D) $T_3T_1T_2$

Open solution

Q17. GATE CSE 2022

Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i, respectively. Consider the schedule S with four transactions.

S : $R_4(x)$, $R_2(x)$, $R_3(x)$, $R_1(y)$, $W_1(y)$, $W_2(x)$, $W_3(y)$, $R_4(y)$

Which one of the following serial schedules is conflict equivalent to S?

(A) $T_1 \rightarrow T_3 \rightarrow T_4 \rightarrow T_2$ (B) $T_1 \rightarrow T_4 \rightarrow T_3 \rightarrow T_2$

(C) $T_4 \rightarrow T_1 \rightarrow T_3 \rightarrow T_2$ (D) $T_3 \rightarrow T_1 \rightarrow T_4 \rightarrow T_2$

Open solution

Q18. GATE CSE 2021 Set 2

Let S be the following schedule of operations of three transactions T_1, T_2 and T_3 in a relational database system:

$R_2(Y)$, $R_1(X)$, $R_3(Z)$, $R_1(Y)$, $W_1(X)$, $R_2(Z)$, $W_2(Y)$, $R_3(X)$, $W_3(Z)$

Consider the statements P and Q below:

P: S is conflict-serializable.

Q: If T_3 commits before T_1 finishes, then S is recoverable.

Which one of the following choices is correct?

(A) P is true and Q is false. (B) Both P and Q are true.

(C) P is false and Q is true. (D) Both P and Q are false.

Open solution

Q19. GATE CSE 2021 Set 1

Let $r_i(z)$ and $w_i(z)$ denote read and write operations respectively on a data item z by a transaction T_i. Consider the following two schedules.

S_1 : $r_1(x)$ $r_1(y)$ $r_2(x)$ $r_2(y)$ $w_2(y)$ $w_1(x)$

S_2 : $r_1(x)$ $r_2(x)$ $r_2(y)$ $w_2(y)$ $r_1(y)$ $w_1(x)$

Which one of the following options is correct?

(A) S_1 is conflict serializable, and S_2 is not conflict serializable.

(B) S_1 is not conflict serializable, and S_2 is conflict serializable.

(C) Both S_1 and S_2 are conflict serializable.

(D) Neither S_1 nor S_2 is conflict serializable.

Open solution

Q20. GATE CSE 2020

Consider a schedule of transactions T_1 and T_2 :

T_1	RA			RC		WD		WB	Commit	
T_2		RB	WB		RD		WC			Commit

Here, RX stands for "Read(X)" and WX stands for "Write(X)". Which one of the following schedules is conflict equivalent to the above schedule?

(A)

T_1				RA	RC	WD	WB		Commit	
T_2	RB	WB	RD					WC		Commit

(B)

T_1	RA	RC	WD	WB					Commit	
T_2					RB	WB	RD	WC		Commit

(C)

T_1	RA	RC	WD				WB		Commit	
T_2				RB	WB	RD		WC		Commit

(D)

T_1					RA	RC	WD	WB	Commit	
T_2	RB	WB	RD	WC						Commit

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Q21. GATE CSE 2016 Set 2

Consider the following database schedule with two transactions, T_1 and T_2

$$S = r_2(X);\ r_1(X);\ r_2(Y);\ w_1(X);\ r_1(Y);\ w_2(X);\ a_1;\ a_2$$

where $r_i(Z)$ denotes a read operation by transaction T_i on a variable Z, $w_i(Z)$ denotes a write operation by T_i on a variable Z and a_i denotes an abort by transaction T_i .

Which one of the following statements about the above schedule is **TRUE**?

(A) S is non-recoverable

(B) S is recoverable, but has a cascading abort

(C) S does not have a cascading abort

(D) S is strict

Open solution

Q22. GATE CSE 2016 Set 1

Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{O_1, \ldots, O_k\}$. This is done in the following manner:
Step 1. T acquires exclusive locks to $O_1, \ldots, O_k$ in increasing order of their addresses.
Step 2. The required operations are performed.
Step 3. All locks are released.

This protocol will

(A) guarantee serializability and deadlock-freedom

(B) guarantee neither serializability nor deadlock-freedom

(C) guarantee serializability but not deadlock-freedom

(D) guarantee deadlock-freedom but not serializability

Open solution

Q23. GATE CSE 2015 Set 3

Consider the following partial Schedule S involving two transactions $T1$ and $T2$. Only the read and the write operations have been shown. The read operation on data item P is denoted by read(P) and the write operation on data item P is denoted by write (P).

Time instance	Transaction-id	
	T1	*T2*
1	*read(A)*	
2	*write(A)*	
3		*read(C)*
4		*write(C)*
5		*read(B)*
6		*write(B)*
7		*read(A)*
8		*commit*
9	*read(B)*	

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Schedule S

Suppose that the transaction $T1$ fails immediately after time instance 9. Which one of the following statements is correct?

(A) $T2$ must be aborted and then both $T1$ and $T2$ must be re-started to ensure transaction atomicity

(B) Schedule S is non-recoverable and cannot ensure transaction atomicity

(C) Only $T2$ must be aborted and then re-started to ensure transaction atomicity

(D) Schedule S is recoverable and can ensure atomicity and nothing else needs to be done

Open solution

Q24. GATE CSE 2015 Set 2

Consider a simple checkpointing protocol and the following set of operations in the log.
(start, $T4$); (write, $T4, y, 2, 3$); (start, $T1$); (commit, $T4$); (write, $T1, z, 5, 7$);
(checkpoint);
(start, $T2$); (write, $T2, x, 1, 9$); (commit, $T2$); (start, $T3$), (write, $T3, z, 7, 2$);

If a crash happens now and the system tries to recovver using both undo and redo operations, what are the contents of the undo list and the redo list?

(A) Undo: $T3, T1$; Redo: $T2$

(B) Undo: $T3, T1$; $\mathrm{Redo}: T2, T4$

(C) Undo: none; Redo: $T2, T4, T3, T1$

(D) Undo: $T3, T1, T4$; Redo : $T2$

Open solution

Q25. GATE CSE 2014 Set 1

Consider the following four schedules due to three transactions (indicated by the subscript) using read and write on a data item x, denoted by r(x) and w(x) respectively. Which one of them is conflict serializable?

(A) $r_1(x); r_2(x); w_1(x); r_3(x); w_2(x)$

(B) $r_2(x); r_1(x);$ w2(x); r3(x); w1(x)

(C) $r_3(x); r_2(x); r_1(x); w_2(x); w_1(x)$

(D) $r_2(x); w_2(x); r_3(x); r_1(x); w_1(x)$

Open solution

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Q26. GATE CSE 2014 Set 2

Open solution

Consider the following schedule S of transactions T1, T2, T3, T4:

T1	T2	T3	T4
	Reads(X)		
		Writes(X) Commit	
Writes(X) Commit			
	Writes(Y) Reads(Z) Commit		
			Reads(X) Reads(Y) Commit

(A) S is conflict-serializable but not recoverable

(B) S is not conflict-serializable but is recoverable

(C) S is both conflict-serializable and recoverable

(D) S is neither conflict-serializable not is it recoverable

Q27. GATE CSE 2014 Set 3

Open solution

Consider the transactions T1, T2, and T3 and the schedules S1 and S2 given below.
T1 : r1 (X) ; r1 (Z) ; w1 (X) ; w1 (Z)

T2 : r2 (Y) ; r2 (Z) ; w2 (Z)

T3 : r3 (Y) ; r3 (X) ; w3 (Y)

S1: r1(X); r3(Y); r3(X); r2(Y); r2(Z); w3(Y); w2(Z); r1(Z); w1(X); w1(Z)

S2: r1(X); r3(Y); r2(Y); r3(X); r1(Z); r2(Z); w3(Y); w1(X); w2(Z); w1(Z)
Which one of the following statements about the schedules is TRUE?

(A) Only **S1** is conflict-serializable.

(B) Only **S2** is conflict-serializable.

(C) Both **S1** and **S2** are conflict-serializable.

(D) Neither **S1** nor **S2** is conflict-serializable.

Q28. GATE CSE 2012

Open solution

Consider the following transactions with data items P and Q initialized to zero:

```
T1 : read (P) ;
     read (Q) ;
     if P = 0 then Q : = Q + 1 ;
     write (Q).

T2 : read (Q) ;
     read (P)
     if Q = 0 then P : = P + 1 ;
     write (P).
```

Any non-serial interleaving of T_1 and T_2 for concurrent execution leads to

(A) a serializable schedule

(B) a schedule that is not conflict serializable

(C) a conflict serializable schedule

(D) a schedule for which precedence graph cannot be drawn

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Q29. GATE CSE 2010

Consider the following schedule for transactions T1, T2 and T3:

T1	T2	T3
Read(X)		
	Read(Y)	
		Read(Y)
	Write(Y)	
Write(X)		
		Write(X)
	Read(X)	
	Write(X)	

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Which one of the schedules below is the correct serialization of the above?

(A) $T1 \rightarrow T3 \rightarrow T2$ (B) $T2 \rightarrow T1 \rightarrow T3$ (C) $T2 \rightarrow T3 \rightarrow T1$ (D) $T3 \rightarrow T1 \rightarrow T2$

Open solution

Q30. GATE CSE 2009

Consider two transactions T_1 and T_2 and four schedules S_1, S_2, S_3, S_4 of T_1 and T_2 as given below:

T_1: $R_1[x]\ W_1[x]\ W_1[y]$

T_2: $R_2[x]\ R_2[y]\ W_2[y]$

S_1: $R_1[x]\ R_2[x]\ R_2[y]\ W_1[x]\ W_1[y]\ W_2[y]$

S_2: $R_1[x]\ R_2[x]\ R_2[y]\ W_1[x]\ W_2[y]\ W_1[y]$

S_3: $R_1[x]\ W_1[x]\ R_2[x]\ W_1[y]\ R_2[y]\ W_2[y]$

S_4: $R_2[x]\ R_2[y]\ R_1[x]\ W_1[x]\ W_1[y]\ W_2[y]$

Which of the above schedules are conflict-serializable?

(A) S_1 and S_2 (B) S_2 and S_3 (C) S_3 only (D) S_4 only

Open solution

Q31. GATE CSE 2009

Let R and S be relational schemes such that R = { a, b, c } and S = { c }. Now consider the following queries on the database:

I. $\pi_{R-S}(r) - \pi_{R-S}\left(\pi_{R-S}(r) \times s - \pi_{R-S,S}(r)\right)$

II. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall u \in s\,(\exists v \in r\,(u = v[S] \wedge t = v\,[R-S]))\}$

III. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall v \in r\,(\exists u \in s\,(u = v[S] \wedge t = v\,[R-S]))\}$

```
IV. Select R.a, R.b
    From R, S
    Where R.c = S.c
```

Which of the above queries are equivalent?

(A) I and II (B) I and III (C) II and IV (D) III and IV

Open solution

Q32. GATE CSE 2008

Consider the following three schedules of transactions T_1, T_2 and T_3.

[Notation: In the following NYO represents the action Y (R for read, W for write) performed by transaction N on object O.]

(S1) 2RA 2WA 3RC 2WB 3WA 3WC 1RA 1RB 1WA 1WB

(S2) 3RC 2RA 2WA 2WB 3WA 1RA 1RB 1WA 1WB 3WC

(S3) 2RZ 3RC 3WA 2WA 2WB 3WC 1RA 1RB 1WA 1WB

Which of the following statements is TRUE?

(A) S1, S2 and S3 are all conflict equivalent to each other

(B) No two of S1, S2 and S3 are conflict equivalent to each other

(C) S2 is conflict equivalent to S3, but not to S1

(D) S1 is conflict equivalent to S2, but not to S3

Open solution

Q33. GATE CSE 2007

Consider the following schedules involving two transactions. Which one of the following statements is TRUE?

S_1: $r_1(X);\ r_1(Y);\ r_2(X);\ w_2(Y);\ w_1(X);$

S_2: $r_1(X);\ r_2(X);\ r_2(Y);\ w_2(Y);\ r_1(Y);\ w_1(X);$

(A) Both S_1 and S_2 are conflict serializable.

(B) S_1 is conflict serializable and S_2 is not conflict serializable.

(C) S_1 is not conflict serializable and S_2 is conflict serializable.

(D) Both S_1 and S_2 are not conflict serializable.

Open solution

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Q34. GATE CSE 2007

Open solution

Consider the following two transactions: T_1 and T_2.

```
T1: read (A);                    T2: read (B);
    read (B);                        read (A);
    if A = 0 then B ← B + 1;         if B ≠ 0 then A ← A - 1;
    write (B);                       write (A);
```

Which of the following schemes, using shared and exclusive locks, satisfy the requirements for strict two phase locking for the above transactions?

(A)

```
S1 : lockS(A);          S2 : lockS(B);
     read (A);               read (B);
     lockS(B);               lockS(A);
     read (B);               read (A);
     if A = 0                if B ≠ 0
     then B ← B + 1;         then A ← A - 1;
     write (B);              write (A);
     commit;                 commit;
     unlock (A);             unlock (B);
     unlock (B);             unlock (A);
```

(B)

```
S1 : lockX(A);          S2 : lockX(B);
     read (A);               read (B);
     lockX(B);               lockX(A);
     read (B);               read (A);
     if A = 0                if B ≠ 0
     then B ← B + 1;         then A ← A - 1;
     write (B);              write (A);
     unlock (A);             unlock (A);
     commit;                 commit;
     unlock (B);             unlock (A);
```

(C)

```
S1 : lockS(A);           S2 : lockS(B);
     read (A);                read (B);
     lockX(B);                lockX(A);
     read (B);                read (A);
     if A = 0                 if B ≠ 0
     then B ← B + 1;          then A ← A - 1;
     write (B);               write (A);
     unlock (A);              unlock (B);
     commit;                  commit;
     unlock (B);              unlock (A);
```

(D)

```
S1 : lockS(A);           S2 : lockS(B);
     read (A);                read (B);
     lockX(B);                lockX(A);
     read (B);                read (A);
     if A = 0                 if B ≠ 0
    then B ← B + 1;          then A ← A - 1;
     write (B);               write (A);
     unlock (A);              unlock (A);
     unlock (B);              unlock (B);
     commit;                  commit;
```

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Q35. GATE CSE 2004

Open solution

Consider the following schedule S of transactions T1 and T2:

T1	T2
Read(A)	
A=A-10	
	Read(A)
	Temp=0.2*A
	Write(A)
	Read(B)
Write(A)	
Read(B)	
B=B+10	
Write(B)	
	B=B+temp
	Write(B)

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Which of the following is **TRUE** about the schedule S?

(A) S is serializable only as T1,T2

(B) S is serializable only as T2,T1

(C) S is serializable both as T1, T2 and T2, T2

(D) S is not serializable either as T1 or as T2

Q36. GATE CSE 2003

Open solution

Consider three data items D1, D2, and D3, and the following execution schedule of transactions T1, T2, and T3. In the diagram, R(D) and W(D) denote the actions reading and writing the data item D respectively.

T1	T2	T3
	R(D3);	
	R(D2);	
	W(D2);	
		R(D2);
		R(D3);
R(D1);		
W(D1);		
		W(D2);
		W(D3);
	R(D1);	
R(D2);		
W(D2);		
	W(D1);	

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Which of the following statements is correct?

(A) The schedule is serializable as T2; T3; T1

(B) The schedule is serializable as T2; T1; T3

(C) The schedule is serializable as T3; T2; T1

(D) The schedule is not serializable

Q37. GATE CSE 1999

Open solution

For the schedule given below, which of the following is correct:

T_1	T_2
Read(A)	
	Read(B)
Write(A)	
	Read(A)
	Write(A)
	Write(B)
Read(B)	
Write(B)	

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(A) This schedule is serialized and can occur in a scheme using 2PL protocol

(B) This schedule is serializable but cannot occur in a scheme using 2PL protocol

(C) This schedule is not serializable but can occur in a scheme using 2PL protocol

(D) This schedule is not serializable and cannot occur in a scheme using 2PL protocol.

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Answer Key

Marks 1

1. **A**	2. **B, C**	3. **C**	4. **A**	5. **A, C, D**
6. **A**	7. **A**	8. **A**	9. **D**	10. **B**
11. **B**	12. **D**	13. **C**	14. **D**	

Marks 2

15. **B**	16. **B, C, D**	17. **A**	18. **A**	19. **B**
20. **A**	21. **C**	22. **A**	23. **B**	24. **A**
25. **D**	26. **C**	27. **A**	28. **B**	29. **A**
30. **B**	31. **A**	32. **D**	33. **C**	34. **C**
35. **D**	36. **D**	37. **D**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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